Neglect-Zero Effects in the Interpretation of Quantifiers and Disjunction

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Several inferences in natural language deviate from classical logic, among them upper-bound inferences (UB) normally analysed as scalar implicatures [8], distributive inferences (DIST) [5] but also certain empty-set effects (ES) in the interpretation of mostly downward-entailing (DE) quantifiers [4]. We present a direct empirical comparison suggesting that diverse tools are needed to capture the processing of these inferences.

(1) a. Some of the squares are black $\rightsquigarrow$ Not all of the squares are black. (UB)
b. Each square is black or white $\rightsquigarrow$ There are black and white squares. (DIST)
c. Less than three squares are white $\rightsquigarrow$ There are white squares. (ES-scope)
d. Less than three squares are white $\rightsquigarrow$ There are squares. (ES-restrictor)

Implicature-based (IMPL) accounts propose a mechanism of comparison and negation of alternatives offering elegant explanations of UB, DIST and ES-scope [8, 5]. ES-restrictor (ES-restr) instead, has been analyzed as a presupposition (PSP) [4, 7]. Recent proposals explain ES-scope [4] and DIST [1, 2] by implementing cognitively motivated mechanisms to model a general tendency to ignore empty configurations (zero-models) in interpretations of natural language sentences [13] called *Neglect-Zero* (NZ). For example a model where all squares are black (■, ■, ■) would be a zero-model for (1-b) and (1-c). This model falsifies DIST and ES-scope. [1, 2] explains these inferences by excluding this model from interpretations of the premises. ES-restr in (1-d) can be explained similarly. To explain ES-scope [4] uses a witness set approach to quantifier interpretation: quantifiers translate into verification algorithms and those with the empty set among their witnesses (ESQ) require a more complex algorithm than other quantifiers: 1) ESQ require an extra rule explicitly dealing with empty scope situations and 2) the explicit representation of negative instances of a predicate (e.g. not white squares). Not performing 1 and 2 in interpretation explains (1-c). [4] treat ES-restr as a PSP (as in [7], but see [12]).

	UB	DIST	ES-scope	ES-restr
[5]	IMPL	IMPL	IMPL	—
[4]	—	—	NZ	PSP
[2]	—	NZ	NZ	NZ

This paper presents an experimental study employing a polar question-answering task allowing for a direct cross-experimental comparison of the inferences in (1). Embedding in questions allowed us to test for the robustness of these inferences. To tease apart effects related to PSP violation from other empty-set effects we introduced three answer options: '*yes, correct*', '*no, incorrect*' or '*odd question*'. To test predicted processing differences between phenomena, we measured reaction times (RT). Previous experiments showed that computation of UBs is cognitively costly ([3, 10], but see [9, 6]), leading to longer RT if the implicature is derived. [1] argued that considering zero-models is cognitively demanding. Previous experiments supported this claim by showing the cognitive cost associated with NZ-violation in DIST (1-b) and ES-scope (1-c) [4, 14].

Experimental Study: 72 German native speakers recruited from Prolific participated in three (sub-)experiments: **Exp. 1 (ESQ)** investigated ES-restrictor and ES-scope in downward entailing (ESQ) vs. upward-entailing (non-ESQ) quantifiers, **Exp. 2 (DIST)** investigated DIST effects in disjunctions embedded under universal quantifiers, and **Exp. 3 (UB)** involved a comparison with scalar *some* to assess RT if participants compute implicatures or choose the literal meaning. (See sample materials below or link to osf). The procedure and RT data preprocessing was similar to [4].

In each trial, on the first screen participants read the question in a self-paced fashion. On the next screen, they saw the images depicting a model for which they had to press one of three arrow keys to provide a '*yes, correct*', '*no, incorrect*' or '*odd question*' response. An experimental session consisted of 140 experimental trials of Exp. 1-3 plus 60 filler trials and typically took 25 min. The answers and cleaned RTs were subjected to mixed effects regression analyses. Answers were analyzed in two separate analyses: 1) *odd question* vs. *yes/no* (GLMER-1) and 2) *yes* vs. *no* (GLMER-2).

Exp. 1: The distribution of answers and RTs of correct and theoretically relevant responses is plotted below (panels A). Across quantifiers, the Empty Restrictor models gave rise to *odd question* responses indicating PSP failure (GLMER-1: main effect of ER-Model vs. rest ($z = 41.13$) in the absence of interactions; contra [12]). We also observed a clear empty-set scope effect, albeit in this case more *no* responses, for ESQs compared to non-ESQs in Empty Scope models relative to the control conditions (GLMER-2: interaction DIRECTION OF ENTAILMENT DOE $\times$ MODEL $z = 6.02$) replicating [4]. These effects were also consistently present in the RT analysis. Reactions to PSP violations for empty restrictors were faster than for any other type of model. For the other conditions, ANSWER POLARITY was included as an additional predictor and an LMER analysis revealed an interaction between POLARITY and DOE (LMER: $t = 4.11$), cf. [11]. Moreover, the predicted three-way interaction was observed due to extraordinary processing difficulty of *less than* in empty scope situations (LMER: $t = -2.23$) (cf. [4]). Unlike for UB (Exp. 3 as well as [3, 10]) pragmatic *no* responses were as fast as logically correct *yes* responses, which constitutes evidence against implicature accounts of ES.

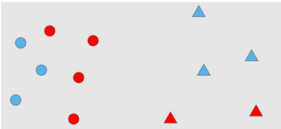
Exp. 2: Panels B show the answers and RTs of the experiment investigating DIST. Again, ES-restr led to *odd question* responses, setting them apart from the other conditions (GLMER-1: $z = 9.69$). Models violating DIST, where one disjunct is emptily supported were accepted less often than true controls (GLMER-2: $z = 4.78$) but considerably more often than false controls (GLMER-2: $z = -9.84$). This difference in answer distributions was also reflected by a significant contrast in RT between the empty condition and the true controls (LMER: $t = -2.06$) suggesting that the suppression of the DIST inference was costly. Also, a general answer polarity effect was observed (LMER: $t = -4.12$).

Answer polarity effects: In the condition of Exp. 3 where UB is violated (e.g all squares are black) *no* responses were on average 879 ms slower than *yes* responses (LMER: $t = 5.75$). To investigate whether the answer polarity effect in Exp. 2 was comparable to the one related to UB, an RT analysis was run including the true and false control conditions for *every* in Exp. 1 as a baseline estimate for a general answer polarity effect [11] in the absence of disjunction (see the RT in panel C). The analysis revealed a significant interaction EXP $\times$ POLARITY for the comparison with the UB of *some* (LMER: $t = 3.41$) but crucially not for DIST inferences (LMER: $t = -0.05$). The latter condition thus only showed an answer polarity effect of the size generally observed for the UE quantifier *every* [11].

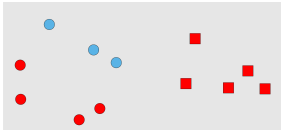
Conclusions: The present study shows that embedded in questions ES-restrictor inferences are very robust, ES-scope effects and DIST inferences are also robust and still frequently present and UB inferences are rare. ES-restrictor inferences were fast and judged as PSP violations. The processing of empty configurations is cognitively demanding for ES-scope and DIST irrespective of answer polarity. In contrast, critical models for UB were only demanding when the inference was derived. This empirical picture supports the usage of multiple tools well tailored to the respective phenomena, such as in [7], [5], [4], [2]. An open question addressed currently in our labs is whether NZ effects observed for quantifiers and DIST can be given a unified account (cf. part of lexical semantics in [4] vs. a pragmatic inference in [1]).

Sample materials (transl. from German; controls not shown; for full set see [link to osf](#))

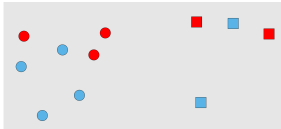
Empty Restrictor



Empty Scope/DIST&UB violation



DIST&UB satisfaction



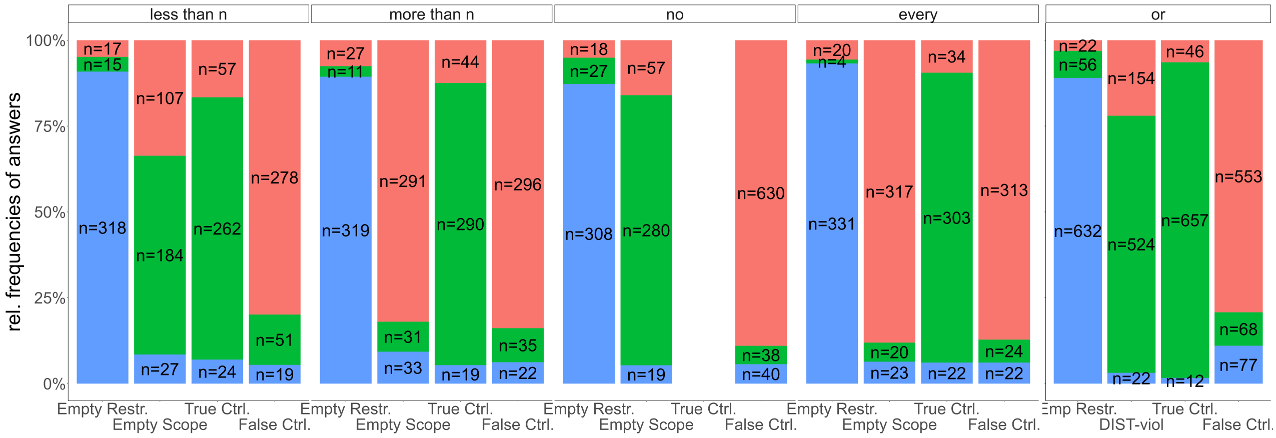
E1	a.	Are less than 3 squares blue?	E2	a.	Is each square red or blue?
	b.	Are more than 3 squares blue?		b.	Is each square blue or red?
	c.	Is no square blue?	E3	a.	Are some squares red?
	d.	Is every square blue?			

Exp. 1 (ESQ): 2 (Q-Type) $\times$ 2 (*DoE*) $\times$ 4 (Model); 80 trials p. part.
Exp. 2 (DIST): 2 (order of disjuncts) $\times$ 4 (Model); 40 trials p. part.
Exp. 3 (UB): 2 conditions; 20 trials p. part.

Response
no, incorrect
yes, correct
odd question

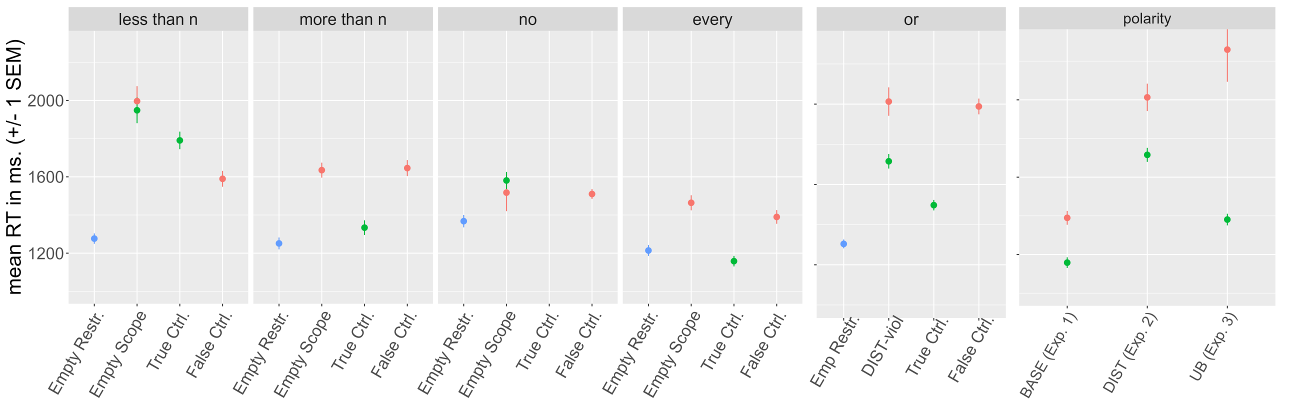
Responses Answer patterns in Exp. 1&2 (Exp. 3: *UB-viol*: 89.3% *yes* vs. 6.5% *no*)

A) Exp. 1 – ESQ B) Exp. 2 – DIST



RT analyses Response RTs: only correct and theoretically relevant responses.

A) Exp. 1 – ESQ B) Exp. 2 – DIST C) Polarity (Exp. 1-3)



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